

Hadoop Installation Guide for Windows

This guide explains how to install Apache Hadoop on Windows in pseudo-distributed mode for Big Data Analytics laboratory work and classroom demonstrations.

Download Links

Apache Hadoop Downloads: <https://hadoop.apache.org/releases.html>

Apache Download Mirror: <https://downloads.apache.org/hadoop/core/>

Oracle JDK Downloads: <https://www.oracle.com/java/technologies/downloads/>

7-Zip Download: <https://www.7-zip.org/>

Prerequisites

1. Install Java JDK 17 or later.
2. Install 7-Zip for extracting Hadoop archives.
3. Ensure at least 8 GB RAM and 10 GB free disk space.

Step 1: Install Java

Download and install JDK. Verify installation using: `java -version`

Step 2: Download Hadoop

Download the Hadoop binary package (.tar.gz) from the Apache website.

Step 3: Extract Hadoop

Extract Hadoop to C:\hadoop using 7-Zip.

Step 4: Configure Environment Variables

Create JAVA_HOME pointing to the JDK folder.

Create HADOOP_HOME pointing to C:\hadoop.

Add %HADOOP_HOME%\bin to the PATH variable.

Step 5: Configure Hadoop Files

Edit core-site.xml, hdfs-site.xml, mapred-site.xml and yarn-site.xml according to pseudo-distributed configuration.

Step 6: Format NameNode

Command: `hdfs namenode -format`

Step 7: Start Hadoop Services

Start HDFS and YARN using `start-dfs.cmd` and `start-yarn.cmd`.

Step 8: Verify Installation

Check services using `jps` command.

Access HDFS UI: <http://localhost:9870>

Access YARN UI: <http://localhost:8088>

Basic Hadoop Commands

```
hdfs dfs -mkdir /input
```

```
hdfs dfs -put file.txt /input
```

```
hdfs dfs -ls /
```

```
hdfs dfs -cat /input/file.txt
```

Learning Outcomes

Understand Hadoop architecture, HDFS, YARN, and basic Hadoop operations.